Eberhard Karls Universität Tübingen | Seminar für Sprachwissenschaft (SfS)
Methods I: Logic, WS 2026/27 | Dr. Kyle Richardson
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Exercise Sheet 3
Propositional Logic 1

Exercise 1: [4 points] Decide whether the following are formulas of propositional logic. For the ones you deem not to be formulas, circle or highlight the problematic part of the expression (no explanation needed).

(a) P_a

(a) _____

(b) $(p \rightarrow \neg q)$

(b) _____

(c) $\neg(P_a \leftrightarrow P_b)$

(c) _____

(d) $(\neg p)$

(d) _____

(e) $((P_a \wedge P_b) \vee \neg P_c)$

(e) _____

(f) $((p \rightarrow q)) \wedge r$

(f) _____

(g) $(P_a \wedge (P_b\neg \vee P_c))$

(g) _____

(h) $\neg((p \vee q) \leftrightarrow (r \rightarrow \neg p))$

(h) _____

Exercise 2: [2 points] Draw a syntactic tree for the following formula.[1]

$$((\neg(p \to \neg q) \wedge r) \leftrightarrow \neg(\neg s \vee p))$$

Exercise 3: [3 points] For each of the following formulas:

1. Indicate whether the formula is **atomic** or **complex**.
2. If it is complex, **circle the main connective** and write its name (e.g., "conjunction," "disjunction") on the line.

(a) P_a

◯ Atomic ◯ Complex

(a) _____

(b) $\neg((p \vee q) \to r)$

◯ Atomic ◯ Complex

(b) _____

(c) $(P_a \wedge \neg P_b) \leftrightarrow \neg(P_c \vee P_d)$

◯ Atomic ◯ Complex

(c) _____

[1] using the same style as in lecture

Exercise 4: [4 points]

(a) Construct a truth table for the following formula. Make sure to include appropriate subformulas to logically build up the final formula.

$$(\neg(p \to q) \vee (p \leftrightarrow q))$$

p	q	$p \to q$	$\neg(p \to q)$	$p \leftrightarrow q$	$(\neg(p \to q) \vee (p \leftrightarrow q))$

(b) Consider the following truth table and highlight or circle the mistakes. Do not correct the table.

$$((p \wedge \neg q) \to (p \leftrightarrow q))$$

p	q	$\neg q$	$p \wedge \neg q$	$p \leftrightarrow q$	$((p \wedge \neg q) \to (p \leftrightarrow q))$
1	1	0	0	1	1
1	0	1	1	0	1
0	1	0	0	0	1
0	0	1	1	1	1

Exercise 5: [5 points] Translating natural language into propositional logic.

(a) For each of the following sentences, decide whether the given formula correctly represents the sentence, using the provided translation key. (**Use Correct or Incorrect labels**) If it is incorrect, write a corrected formula.

i. 'If the alarm rings, the security guard calls the police.'
Translation Key:
p - The alarm rings.
q - The security guard calls the police.
Formula: $(q \rightarrow p)$

i. _____

Corrected Formula (if needed):

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ii. 'The professor explained the problem, but the students did not understand it.'
Translation Key:
P_a - The professor explained the problem.
P_b - The students understood the problem.
Formula: $(P_a \vee \neg P_b)$

ii. _____

Corrected Formula (if needed):

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iii. 'The experiment is successful if and only if the participants follow the instructions.'
Translation Key:
p - The experiment is successful.
q - The participants follow the instructions.
Formula: $(p \leftrightarrow q)$

iii. _____

Corrected Formula (if needed):

(b) For each of the following natural language sentences, create a translation key and write a logical form[2] that represents its logical structure.

i. 'The researcher collected the data and analyzed the results, but did not publish the findings.'
Translation Key:

Formula:

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[2] A well-formed formula of propositional logic.

ii. 'If the experiment fails, two things will happen: the researchers will either repeat the experiment or abandon the project, and they will not publish the results.'
Translation Key:

Formula: